



DATA DRIVEN DEFENSE

MACHINE LEARNING
IN NETWORK TRAFFIC
ANALYSIS



What is Network Traffic Analysis?



NTA is a method of monitoring network availability and activity to identify anomalies, including security and operational issues



Helps minimize attack surface, reduce response times, meet compliance requirements



Detects network vulnerabilities and potential attacks (malware, breaches, DDoS, MITM)

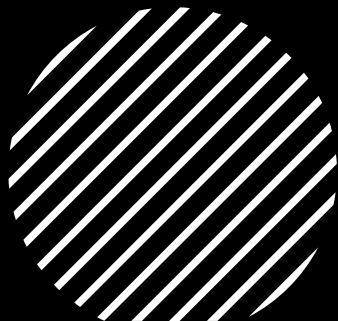




What about Firewalls and Logs?

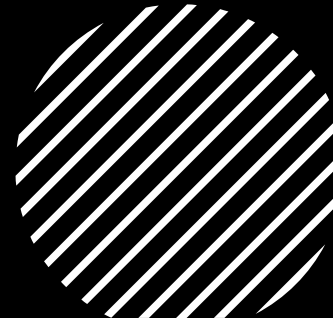
NTA provides an organization with more visibility into threats on their networks beyond the endpoint

With the increasing use of mobile devices, IoT, industrial machinery, you need something with more intelligence than just the logs from firewalls





(Industrial) + Internet of Things



“By the end of 2024, there are projected to be more than **207 billion devices connected** to the worldwide network of tools, toys, devices and appliances that make up the Internet of Things (IoT)”
(Forbes, 2023)

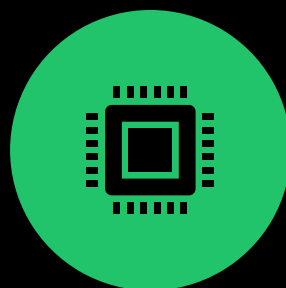
“The value of the IoT healthcare market has been predicted to grow to around \$150 billion this year, on its way to reaching a valuation of **\$289 billion by 2028**” (Forbes, 2023)



IoT and IIoT Security



Devices often operate autonomously, NTA helps protect these networks from attacks targeting vulnerable devices



These devices may lack strong internal security, so analyzing traffic patterns is one of the most effective ways to spot potential threats



RT analysis of network traffic allows rapid detection and response to threats



Critical given the minimal latency tolerance in many IIoT applications (manufacturing, healthcare)





Intrusion Detection Systems (IDS)

IDS are a security solution designed to monitor network or system activities for malicious activities or policy violations

IDS identifies and flags suspicious behaviors or potential threats, alerting network administrators to unauthorized access, attacks, or malware

Some IDS are purely detection-based, they alert admins to take action, other systems are paired with Intrusion Prevention Systems (IPS) to automatically block or mitigate threats





Why ML in IDS?



ML techniques are particularly valuable for anomaly-based IDS, as ML algorithms can learn from large datasets and adapt to new patterns



This enables IDS to detect sophisticated or evolving threats that may not match known signatures

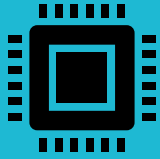


ML enables IDS to go beyond fixed rules by understanding network behaviors, they allow the detection of subtle changes that might indicate slow and stealthy attacks





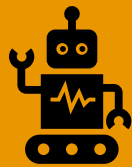
Edge-IIoTset Dataset (2022)



Realistic, comprehensive cybersecurity dataset tailored for IoT and Industrial IoT applications



Created to support research in network traffic analysis, particularly in centralized and federated learning setups for intrusion detection and anomaly detection in IoT/IIoT networks



Collected from a sophisticated seven-layer testbed including more than 10 IoT devices, IIoT-based Modbus flows, 14 IoT and IIoT protocol-related attacks

- Credit where credit is due!



Mohamed
Ferrag



Othmane
Friha



Djallel
Hamouda



Leandros
Maglaras

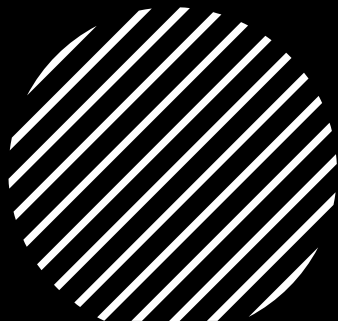


Helge
Janicke





Proposed Testbed (Seven Layers)



Cloud Computing

Network Functions Virtualization

Blockchain Network

Fog Computing

Software-Defined Networking

Edge Computing

IoT and IIoT Perception



Let's Connect

- <https://www.laplacef.me>
- <https://github.com/laplacef>

